14.2.1 Classification of Bases

(a) Based on their Acidity

Monoacidic Base: It is a base that ionises in water to give one hydroxide ion per molecule. Example: NaOH, KOH

Diacidic Base: It is a base that ionises in water to give two hydroxide ions per molecule. **Example:** $Ca(OH)_2$. $Mg(OH)_2$

Triacidic Base: It is a base that ionises in water to give three hydroxide ions per molecule. **Example:** $Al(OH)_3$, $Fe(OH)_3$

(b) Based on concentration

Concentrated Alkali: It is an alkali having a relatively high percentage of alkali in its aqueous solution.

Dilute Alkali: It is an alkali having a relatively low percentage of alkali in its aqueous solution.

(c) Based on Ionisation

Strong Bases: These are bases which ionise completely in aqueous solution.
Example: NaOH, KOH

Weak Bases: These are bases that ionise partially in aqueous solution.
Example: NH_4OH, $Ca(OH)_2$

DO YOU KNOW?

The term acidity is used for base, which means the number of replaceable hydroxyl groups present in one molecule of a base.

14.2.2 Properties of Bases

a) They have bitter taste.
b) Their aqueous solutions have soapy touch.
c) They turn red litmus blue.
d) Their aqueous solutions conduct electricity.
e) Bases react with metals to form salt with the liberation of hydrogen gas.

$$Zn + 2\ NaOH \rightarrow Na_2ZnO_2 + H_2\uparrow$$

f) Bases react with non-metallic oxides to produce salt and water. Since this is similar to the reaction between a base and an acid, we can conclude that non-metallic oxides are acidic in nature.

$$Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$$

g) Bases react with acids to form salt and water.

$$KOH + HCl \rightarrow KCl + H_2O$$

The above reaction between a base and an acid is known as Neutralisation reaction.

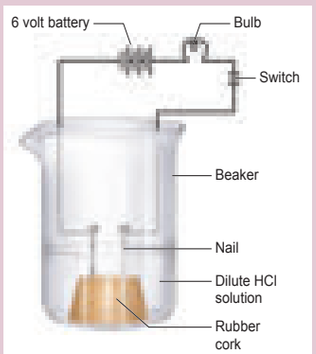
h) On heating with ammonium salts, bases give ammonia gas.

$$NaOH + NH_4Cl \rightarrow NaCl + H_2O + NH_3\uparrow$$

DO YOU KNOW?

Few metals do not react with sodium hydroxide. Example: Cu, Ag, Cr

Activity 2



Take solutions of hydrochloric acid or sulphuric acid. Fix two nails on a cork and place the cork in a 100 ml beaker. Connect the nails to the two terminals of a 6V battery through a bulb and a switch as shown in Figure. Now pour some dilute HCl in the beaker and switch on the current. Repeat the activity with dilute sulphuric acid, glucose and alcohol solutions.What do you observe now? Does the bulb glow in all cases?

In the above activity you can observe that the bulb will start glowing only in the case of acids. But, you will observe that glucose and alcohol solution do not conduct electricity. Glowing of the bulb indicates that there is a flow of electric current through the solution. The electric current is carried through the solution by ions. Repeat the same activity using alkalis such as sodium hydroxide and calcium hydroxide.

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